A Literacy-Free, Report-Free Eye Exam on a Consumer EEG Headband and VR Headset: Toward Objective Vision Screening by Community Health Workers

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***Abstract*—At least a billion people have a vision impairment that could be corrected, and the bottleneck is the screen: an eye test needs a literate, cooperating subject and a trained examiner. We describe an exam that asks nothing of the subject but to look at a dot with one eye covered, on a Muse headband with one electrode added over the visual cortex and a Meta Quest 3S running the stimulus in its browser, under 600 US dollars together. Acuity, contrast, field and colour come from the steady-state visual evoked potential, an exact test per step and the sweep-VEP threshold rule. A hardware-free phantom with set cut-offs and a dead sector passes all twelve checks of a full-stack gate. The first sitting on a person fitted no threshold because two electrodes had come loose unnoticed; it produced the contact check a device in a health worker's hands must have. Nothing here is a diagnosis.**

***Index Terms*—mobile health; visual screening; SSVEP; sweep VEP; wearable sensors; consumer electronics; community health workers.**

I. The Screen, Not the Lens, Is the Bottleneck

The WHO counts at least 2.2 billion people with a vision impairment, at least a billion of them addressable, uncorrected refractive error first among the causes [1]. A letter chart needs a subject who can read it in the examiner's language and an examiner to judge the answers. The pattern that has carried other screens to community health workers is a consumer device, a worker, and software that checks itself [2], [3]. The steady-state visual evoked potential (SSVEP) gives an objective acuity test with a clinical pedigree, the sweep VEP built for infants, who cannot report [4], and a contrast threshold [5]; it removes literacy, language and examiner bias while bringing biases of its own in hair and headband fit [6].

II. The Rig and the Exam

Mann et al. turned the eye into a camera with the SSVEP [7]; we rebuilt that rig on a Muse headband with a cup electrode at Oz, the phone streaming raw EEG as the hub in the wireless-sensing architecture iCalm set out [8], and a Quest 3S whose browser renders the stimulus (Figs. 1, 2). Each test is ON/OFF blocks of flicker (4 s, two pairs per step, 12 Hz on the 72 Hz display): a grating appearing and disappearing on a grey of the same mean luminance with bars from 40 to 3.5 arcmin (acuity) or contrast from 1 to 0.03 (contrast), one sector of a 4×4 grid at a time (field), the patch in red, green or

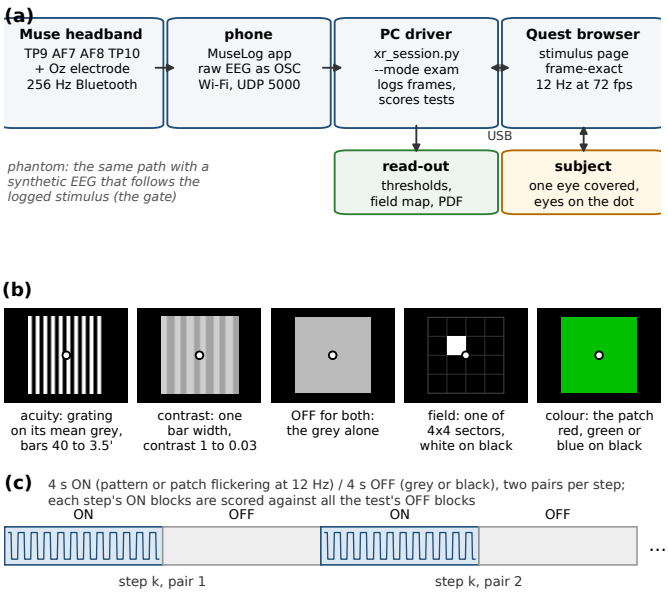


Fig. 1. (a) Scalp to read-out: the headband streams through the phone to the PC, which drives the page in the headset's browser, logs every frame and scores each test; the phantom replaces the first two boxes with a synthetic EEG that follows the logged stimulus. (b) The four stimuli and the OFF screen of the grating tests. (c) One step's blocks.

blue (colour). Each step's ON blocks are scored against all the test's OFF blocks by the line power at the flicker frequency and its double, with an exact permutation test; a channel above 60 μV rms through a test is set aside. Thresholds follow the sweep-VEP rule: a line through the limb from the step below the finest responding one up the rise, extrapolated to zero. Seven minutes per eye.

III. Validation Without a Patient

A device for a worker with no eye clinic behind them must be testable with no patient in front of it. A phantom subject reads the stimulus log as it is written and streams synthetic EEG whose SSVEP follows what is on screen, through a logistic visibility in bar width and contrast with a cut-off per eye, dead sectors and colour gains. The gate dials in a right eye at 8 arcmin and contrast 0.08 with one dead sector, a left eye twice as coarse and whole, gains 0.7/1/0.4 for red/green/blue,

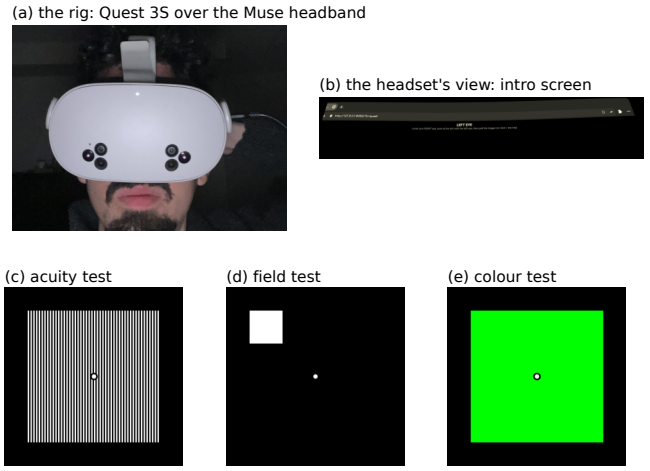


Fig. 2. (a) The Quest 3S over the Muse headband. (b) The headset's own view of one eye's intro screen. (c)–(e) The page during an ON frame of the acuity, field and colour tests.

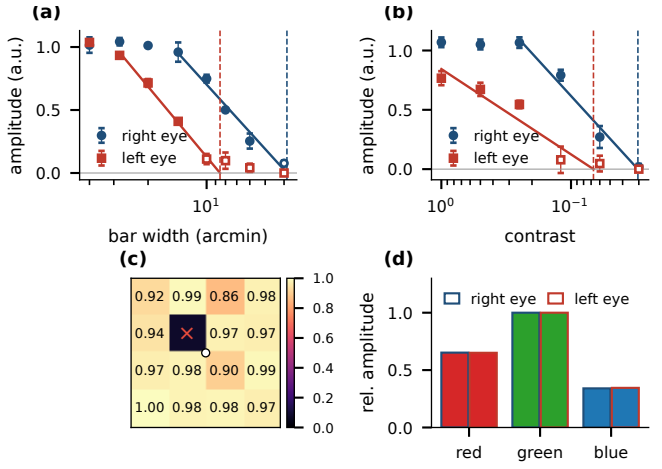


Fig. 3. The gate's read-out. (a) Acuity and (b) contrast per phantom eye (right circles, left squares; open: $p \geq 0.05$), fitted limbs and their zero crossings. (c) The right eye's field, the cross its dead sector. (d) Colour planes.

and asks twelve things: every stage delivered, each threshold inside the band where the phantom's visibility runs from 0.03 to 0.6, the left eye at least $1.3\times$ coarser, exactly the dead sector named, green $\geq$ red $>$ blue. All twelve pass (Fig. 3): right eye 3.9 arcmin and 0.031 with sector six silent, left eye 7.8 arcmin and 0.067, response phase within 42 to 47° in every responding sector.

IV. The First Sitting

One of the authors ran the right eye with the left covered, then the left, at 12 Hz. No threshold could be fitted: the only clear response was the full-patch colour flicker (12 Hz line on Oz, $\log_{10}$ SNR $+0.75$ ON against -0.22 OFF); the gratings gave little, the sectors none. Both ear electrodes had sat at 100 to 280 μV rms all sitting, loose under the headset strap, and the session's signal check had passed them because it was written for electrodes at the amplifier rails; Oz held for the right eye and drifted after: the quiet failure, not the wrong number, that a consumer wearable brings to the field [3]. The exam now measures every channel before each eye and, when one is above 60 μV, tells the subject on the headset which electrode to press in and waits; the read-out names any channel set aside.

V. What the Same Recording Carries

A headband records more than the visual response [9]. In companion sittings on the same rig (in preparation) the forehead electrodes gave the blink rate, which stops while a patch flickers, bursts when it goes dark, and doubled between two sittings of one evening, an old marker of fatigue and eye strain [10]. That is the hook for a just-in-time adaptive intervention [11]: the device that has just measured an eye can notice the eyes are tired and say so. We call it an eye-strain estimate against the subject's own earlier minutes, not detection and not emotion.

VI. Toward the Field

Established: the apparatus, an exact test per step, a threshold rule from the sweep VEP, and a phantom gate that catches failures without a volunteer. Not established: any measurement on a person. Open before a second sitting: the headset's pixels per degree (the arcmin scale) and the flicker rate of the pattern tests. Then a blur study with trial lenses against a letter chart, a normative cohort, and a worker-facing flow in which the device checks its own contact and the report is one picture.

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